LanguageName: PLPDI Mini-Project

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Repository: `https://github.com/DavidTilman/plpdi-mini-project`
Commit: `TODO`

1 Informal Description

2 Literature Review

3 Formal Syntax and Semantics

3.1 Syntax

$\oplus$	::=	$+ \mid - \mid * \mid / \mid \% \mid < \mid <= \mid == \mid != \mid >= \mid > \mid \mid\mid \mid \&\&$	binary operators
τ	::=	`Nat` $\mid$ `Int` $\mid$ `Real` $\mid$ `Bool`	base types
	$\mid$	$(\tau_1, \ldots, \tau_n) \to \tau$	function type
	$\mid$	R	record type
e	::=	n	literal
	$\mid$	`True` $\mid$ `False`	bool literal
	$\mid$	x	variable
	$\mid$	$e_1 \oplus e_2$	binary operation
	$\mid$	$e.f$	field access
	$\mid$	$e(e_1, \ldots, e_n)$	application
	$\mid$	$R(e_1, \ldots, e_n)$	record construction
	$\mid$	`Lambda` $(x_1 : \tau_1, \ldots, x_n : \tau_n) : \tau \Rightarrow e$	lambda
	$\mid$	$e \ ? \ e_1 : e_2$	if expression
b	::=	$\{ \, s_1; \ \ldots; \ s_n; \, \}$	block ($n \geq 0$)
s	::=	`Var` $x : \tau = e$	declaration
	$\mid$	$x := e$	assignment
	$\mid$	`While` (e) b	loop
	$\mid$	`If` (e) b	conditional
	$\mid$	`If` (e) b_1 `Else` b_2	conditional with else
	$\mid$	`Ret` e	return
d	::=	`Record` R $(f_1 : \tau_1, \ldots, f_n : \tau_n)$	record declaration
	$\mid$	`Func` F $(x_1 : \tau_1, \ldots, x_n : \tau_n) : \tau = b$	function declaration
c	::=	$d \mid s$	top-level item
p	::=	$c_1; \ \ldots; \ c_n;$	program

`If` (e) b abbreviates `If` (e) b `Else` $\{ \}$.

3.2 Type System

3.3 Operational Semantics

4 Implementation

4.1 Architecture

4.2 Correspondence with the Formal Semantics

5 Distinctive-Feature Example Program

6 Reflection on AI and Collaboration

A Remaining Operational Semantics Rules